



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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UNIT TEST - 1, JULY - 2024

Time Allowed : 1.30 Hours]

MATHEMATICS

[Max. Marks : 50

PART - A**I. Choose the correct Answer.****7x1=7**

1. If $n(A \times B) = 6$ and $A = \{1, 3\}$ then $n(B)$ is ----
 (a) 1 (b) 2 (c) 3 (d) 6
2. If the ordered pairs $(a+2, 4)$ and $(5, 2a+b)$ are equal then (a, b) is
 (a) $(2, -2)$ (b) $(5, 1)$ (c) $(2, 3)$ (d) $(3, -2)$
3. If $f(x) = 2x^2$ and $g(x) = \frac{1}{3x}$, then $f \circ g$ is
 (a) $\frac{3}{2x^2}$ (b) $\frac{2}{3x^2}$ (c) $\frac{2}{9x^2}$ (d) $\frac{1}{6x^2}$
4. If the HCF of 65 and 117 is expressible in the form of $65m - 117$, then the value of m is
 (a) 4 (b) 2 (c) 1 (d) 3
5. Given $F_1 = 1$, $F_2 = 3$ and $F_n = F_{n-1} + F_{n-2}$ then F_5 is
 (a) 3 (b) 5 (c) 8 (d) 11
6. $7^{4k} \equiv \text{---} \pmod{100}$
 (a) 1 (b) 2 (c) 3 (d) 4
7. In $\triangle LMN$, $\angle L = 60^\circ$, $\angle M = 50^\circ$. If $\triangle LMN \sim \triangle PQR$ then the value of $\angle R$ is
 (a) 40° (b) 70° (c) 30° (d) 110°

PART - B**II. Answer any five questions only. [Q.No. 14 is compulsory].****5x2=10**

8. Let $A = \{1, 2, 3\}$ and $B = \{x | x \text{ is a prime number less than } 10\}$ Find $A \times B$ and $B \times A$.
9. Let $f(x) = 2x + 5$. If $x \neq 0$ then find $\frac{f(x+2) - f(2)}{x}$
10. Find the first four terms of the sequences whose n^{th} term is given by $a_n = 2n^2 - 6$
11. If $13824 = 2^a \times 3^b$ then find a and b
12. Solve: $5x \equiv 4 \pmod{6}$
13. Find all positive integers, when divided by 3 leaves remainder 2.

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14. Is $7 \times 5 \times 3 \times 2 + 3$ a Composite Number? Justify your answer.

PART - C

III. Answer any 5. Q.No. 21 is compulsory.

5x5=25

15. Let $A = \{x \in W \mid x < 2\}$, $B = \{x \in N \mid 1 < x \leq 4\}$ and $C = \{3, 5\}$

Verify that $A \times (B \cup C) = (A \times B) \cup (A \times C)$.

16. Let $f : A \rightarrow B$ be a function defined by $f(x) = \frac{x}{2} - 1$, where $A = \{2, 4, 6, 10, 12\}$ $B = \{0, 1, 2, 3, 4, 5, 9\}$

Represent f by i) Set of ordered pairs ii) a table iii) an arrow diagram iv) a graph

17. Find the HCF of 396, 504, 636.

18. Find the number of Integer solutions of $3x \equiv 1 \pmod{15}$

19. Find a_8 and a_{15} whose n^{th} term is $a_n = \begin{cases} \frac{n^2 - 1}{n + 3} & ; n \text{ is even } n \in N \\ \frac{n^2}{2n + 1} & ; n \text{ is odd, } n \in N \end{cases}$

20. Find the value of P_1, P_2, P_3, P_4 and x_1, x_2, x_3, x_4 if $P_1^{x_1} \times P_2^{x_2} \times P_3^{x_3} \times P_4^{x_4} = 113400$ where P_1, P_2, P_3, P_4 are primes in ascending order and x_1, x_2, x_3, x_4 are integers.

21. If $f(x) = 2x+3$, $g(x) = 1 - 2x$ and $h(x) = 3x$ Prove that $fo(goh) = (fog)oh$

PART - D

IV. Answer Any One of the following.

1x8=8

22. Construct a triangle similar to a given triangle PQR with its sides equal to $\frac{3}{5}$ of the corresponding sides of the triangle PQR (Scale factor $\frac{3}{5} < 1$)

23. Construct a triangle similar to a given triangle ABC with its sides equal to $\frac{6}{5}$ of the corresponding sides of the triangle ABC (Scale factor $\frac{6}{5} > 1$)